

Back to Square One

Foundations of AI

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26 September 2026

ANSWER KEY

How this works

- Nine questions. Some are multiple choice: put a hand up.
- The rest you **write down yourself**, on paper or on a laptop, and then we fill them in together on the screen.
- Everything on these slides was on a slide today.

A question the room gets wrong is worth five if it gets right, so guess out loud and be wrong cheerfully. That has been the whole day.

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Section 1

The loop

1. Write the algorithm

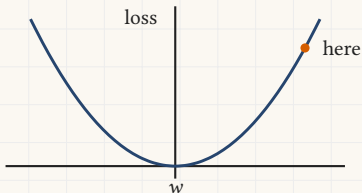
Block 1, one house. x is its size in square metres, y the price it sold for, and w the price per square metre we are looking for. The model is $y = w * x$.

```
def fit(x, y, w, alpha):  
    l = y - w * x      # the verdict: y minus the estimate  
    while abs(l) > 0.5: # while more than half a franc out  
        w = w + alpha * l # the update: move w the way the miss points  
        l = y - w * x     # score the new guess  
    return w
```

On a 92 square metre house that sold for 2.39 million, starting from 20 000 francs per square metre with $\alpha = 0.01$, this returns 25 980 after eight rounds.

2. The tilt, on the side of the valley

The curve is the loss. The dot is where the guess is now. What is the gradient there?

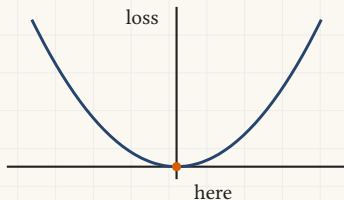


- A) about -12
- B) exactly 0
- C) about 0.1
- D) about $+12$

D Uphill to the right, so the tilt is positive, and the wall is steep, so it is large. The step is that number times alpha, taken the other way.

3. The tilt, at the bottom

Same curve. The dot has moved.



- A) about -12
- B) exactly 0
- C) about $+0.5$
- D) about $+12$

B Flat ground: the tilt is zero, the step is zero, and the grasshopper stops. Careful, though. It stops on ANY flat spot, lowest or not.

4. Which of these can draw any shape at all?

Given enough of them. More than one answer is right.

- A) Neural networks
- B) Straight lines
- C) Polynomials
- D) Loss functions

A and C

Both. A network with enough bends can approximate any shape, and so can a polynomial of high enough degree. Neither promise says how many bends or what degree, which is the catch.

5. The network inside a language model

You type half a sentence and press enter. What does the network itself hand back?

- A) A word
- B) A matrix of words
- C) One probability for every word it knows
- D) The word with the highest score

C It never hands back a word. It hands back a chance for each of the fifty thousand or so words it knows, and something outside the network picks one. D is what that something does next.

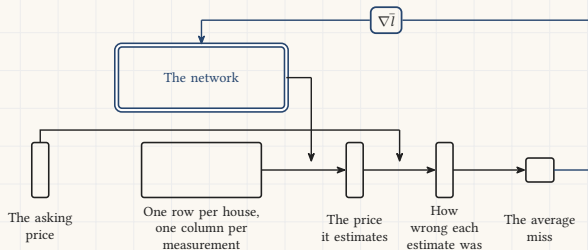
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Section 2

Name the boxes

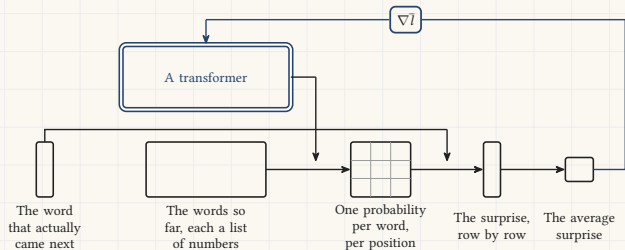
6. The same five boxes: house prices

Every application today was this one picture. Here it is a network that predicts house prices.



7. The same five boxes: a language model

Nothing about the picture changes. Only what is in the boxes. One row is a piece of text with its next word cut off.



8. That wider box, on its own

Two short texts, *the cat sat* and *the dog ran*. The five words along the top are everything this toy model knows.

<i>the</i>	<i>cat</i>	<i>sat</i>	<i>dog</i>	<i>ran</i>	
0	1	0	0	0	after <i>the</i> , the next word was <i>cat</i>
0	0	1	0	0	after <i>the cat</i> , it was <i>sat</i>
0	0	0	0	1	after <i>the dog</i> , it was <i>ran</i>

A **row** is one position in the text, so one training example. A **column** is one word of the vocabulary. A perfect model puts **1** under the word that actually came next and **0** everywhere else. A real one spreads that 1 out, and how thinly it spreads it is the loss.

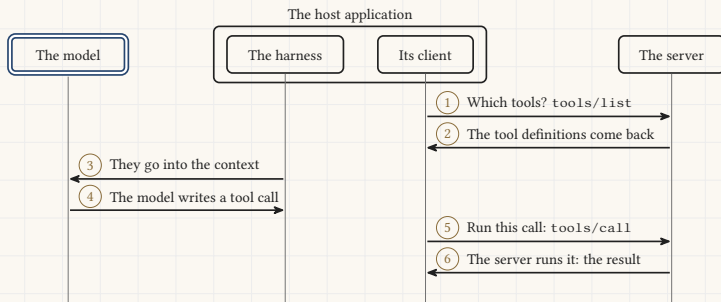
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Section 3

The agent

9. What travels along each arrow?

The four actors of this afternoon, and the six messages between them, in order.



That is the day

- A guess, a measurement of how wrong it was, and a correction. Nine o'clock to five.
- The same five boxes behind a house price and behind a sentence.
- The only honest number comes from data the model never saw.
- A model writes text. A harness does things. Govern the harness.

Your project is on the course page: about an hour, in the browser, and it builds the house price network you met at question 6. Thank you for a Saturday.